TEREK JOHNSON

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EDUCATION

Master of Science, Computer Science, University of California, Riverside — Expected June 2027

Bachelor of Science, Computer Science, University of California, Riverside — GPA 3.51, June 2026

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Embedded Systems, Operating Systems, Computer Architecture

WORK EXPERIENCE

Project Data Systems Engineering Intern — June 2026 – Aug 2026
NASA Jet Propulsion Laboratory (JPL) — *Pasadena, CA*

- Wrote specifications in Kaitai Struct from Deep Space Network standards to parse/serialize nested binary telemetry previously unsupported by JPL's legacy Telemetry Tracking & Command (TTC) software.
- Built a maintainable, lightweight parsing library for JPL's project data system engineers.

Transfer Student Peer Mentor — Sep 2025 – June 2026
Marlan & Rosemary Bourns College of Engineering, University of California, Riverside — *Riverside, CA*

- Directed marketing campaigns, growing non-follower engagement by 8%, advised transfer/prospective students on academic/career growth, managed study spaces, and represented the college at outreach events.

TECHNICAL PROJECTS

Artificial Horizon from Orientation & Yaw (AHOY!) — May 2026
Flight instrument utilizing parallel processing for real-time sensor fusion and SPI/UDP communication protocols.

- Designed FPGA-based flight attitude display system interfacing with IMU via complementary filter for real-time pitch and roll orientation estimation.
- Developed communication pipeline from FPGA to Raspberry Pi controlling LCD and transmitting to a wireless Python ground station.

Single-Axis Solar Tracker & Wireless Transceiver Network — Jan 2026 – Mar 2026
Optimized the angle of solar panels for a solar boat and transmitted data for a statewide collegiate competition.

- Engineered a solar tracker using lux sensors, an Arduino Nano, and an H-bridge-driven DC motor to orient a solar panel toward peak sunlight for a maximum collection.
- Built a wireless transceiver system using nRF antennas, integrating a GPS module, an IMU, a DC power sensor, and lux sensors to transmit telemetry from the lake to a shore-side station, enabling live monitoring.
- Coordinated design and integration across mechanical, electrical, and software subteams.

Machine Learning Model Analysis on Titanic Dataset — Dec 2025
Compared different nearest-neighbor AI algorithms to create a model to predict survival of passengers.

- Applied one-nearest-neighbor forward/backward feature-elimination to identify the optimal feature subset for a passenger survival-prediction model, then diagnosed a critical training error improving accuracy by 30%.

RESEARCH EXPERIENCE

Undergraduate Research Assistant — June 2024 – Sep 2024
Collaborative Intelligent Systems Lab, University of California, Riverside — *Riverside, CA*

- Collected and synchronized a multi-modal LiDAR/camera dataset with a team of eight, using instrumented vehicles and city sensors to support autonomous-driving machine learning (ML) pipelines.
- Evaluated data quality and debugged sensor drivers, validating the dataset for training/benchmarking ML models, presented findings at the 2024 Bourns College of Engineering Summer Research Symposium.

SKILLS & HONORS

Languages: C++, C, C#, Python, Verilog/SystemVerilog, JavaScript, Java

Frameworks & Tools: React, HTML, CSS, NodeJS, Docker, MongoDB, Git, CI/CD

Honors: University Regents Scholar (2024) • Dean's Honor List • Southern California Edison Scholarship (2025)